Documentation and operational readiness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete runbook and rehearsal-evidence record.

Keep versions, fixture, checks, owner, fallback, compatibility and actual operator observations together.

Keep one habit: Recall → W1–W3/W5 → independent readiness-gap case → runbook project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.
- Without opening the old lesson, explain one key idea from Red team exercises. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Performance and cost experiments. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Incident investigation. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Operational documentation tells another person how to run, verify, observe and recover the service. A successful developer demo is not readiness.

- A **runbook** names versions, required non-secret configuration, secret key names, **health check**s, release/rollback dependencies, alerts and owners. Test it with a fresh operator and fictional fixtures.

Red team exercises

> **My remembered idea, example and correction**

Performance and cost experiments

> **My remembered idea, example and correction**

Incident investigation

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Operational documentation should let another person run permitted setup, verify expected behavior, observe failures and review compatible recovery. Main setup-only note omits data version and recovery, so add code/config/corpus/procedure/schema/fixture references, non-secret config, secret names only, health checks, alerts and owners.

  Complete paper fiction: toy-r2/cfg 2/c1/p2/schema 1/F1; ENV classroom-mock; DATA_VERSION c1; F1 approved B12 with source_available true. Healthy expected answer B12 cites F1/c1. Unavailable variant returns SOURCE_UNAVAILABLE and needs_review, no guessed room. Inspect only approved read-only configuration/availability and prepare a minimized packet for the symbolic maintainer/authorized route; no real contact occurs.

  Proposed toy-r1 recovery requires current schema 1/c1; schema 2 blocks. No actual server, diagnostic connection, secrets or rollback are used. Another operator rehearsal must be actually performed and recorded; otherwise label self-trace/not run. Picture removal of owner/route leaves escalation incomplete. A clear/rehearsed runbook provides scoped evidence, not readiness for every incident.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Operational documentation should let another person run permitted setup, verify expected behavior, observe failures and review compatible recovery. Main setup-only note omits data version and recovery, so add code/config/corpus/procedure/schema/fixture references, non-secret config, secret names only, health checks, alerts and owners.

  Complete paper fiction: toy-r2/cfg 2/c1/p2/schema 1/F1; ENV classroom-mock; DATA_VERSION c1; F1 approved B12 with source_available true. Healthy expected answer B12 cites F1/c1. Unavailable variant returns SOURCE_UNAVAILABLE and needs_review, no guessed room. Inspect only approved read-only configuration/availability and prepare a minimized packet for the symbolic maintainer/authorized route; no real contact occurs.

  Proposed toy-r1 recovery requires current schema 1/c1; schema 2 blocks. No actual server, diagnostic connection, secrets or rollback are used. Another operator rehearsal must be actually performed and recorded; otherwise label self-trace/not run. Picture removal of owner/route leaves escalation incomplete. A clear/rehearsed runbook provides scoped evidence, not readiness for every incident.

- **Runbook** says start server, but omits data version and recovery steps. Another operator cannot reproduce a result. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Operational documentation should let another person run permitted setup, verify expected behavior, observe failures and review compatible recovery. Main setup-only note omits data version and recovery, so add code/config/corpus/procedure/schema/fixture references, non-secret config, secret names only, health checks, alerts and owners.

  Complete paper fiction: toy-r2/cfg 2/c1/p2/schema 1/F1; ENV classroom-mock; DATA_VERSION c1; F1 approved B12 with source_available true. Healthy expected answer B12 cites F1/c1. Unavailable variant returns SOURCE_UNAVAILABLE and needs_review, no guessed room. Inspect only approved read-only configuration/availability and prepare a minimized packet for the symbolic maintainer/authorized route; no real contact occurs.

  Proposed toy-r1 recovery requires current schema 1/c1; schema 2 blocks. No actual server, diagnostic connection, secrets or rollback are used. Another operator rehearsal must be actually performed and recorded; otherwise label self-trace/not run. Picture removal of owner/route leaves escalation incomplete. A clear/rehearsed runbook provides scoped evidence, not readiness for every incident.

- A report omits this qualification: Documentation readiness requires a tested replay and **operational owner**ship, not merely complete-looking prose.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Operational documentation should let another person run permitted setup, verify expected behavior, observe failures and review compatible recovery. Main setup-only note omits data version and recovery, so add code/config/corpus/procedure/schema/fixture references, non-secret config, secret names only, health checks, alerts and owners.

  Complete paper fiction: toy-r2/cfg 2/c1/p2/schema 1/F1; ENV classroom-mock; DATA_VERSION c1; F1 approved B12 with source_available true. Healthy expected answer B12 cites F1/c1. Unavailable variant returns SOURCE_UNAVAILABLE and needs_review, no guessed room. Inspect only approved read-only configuration/availability and prepare a minimized packet for the symbolic maintainer/authorized route; no real contact occurs.

  Proposed toy-r1 recovery requires current schema 1/c1; schema 2 blocks. No actual server, diagnostic connection, secrets or rollback are used. Another operator rehearsal must be actually performed and recorded; otherwise label self-trace/not run. Picture removal of owner/route leaves escalation incomplete. A clear/rehearsed runbook provides scoped evidence, not readiness for every incident.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Documentation includes all setup steps but no incident owner or compatible rollback. Identify readiness gaps.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled documentation and operational readiness **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of documentation and operational readiness.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Documentation and operational readiness. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Use the worked case for Documentation and operational readiness. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Documentation includes all setup steps but no incident owner or compatible rollback. Identify readiness gaps.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

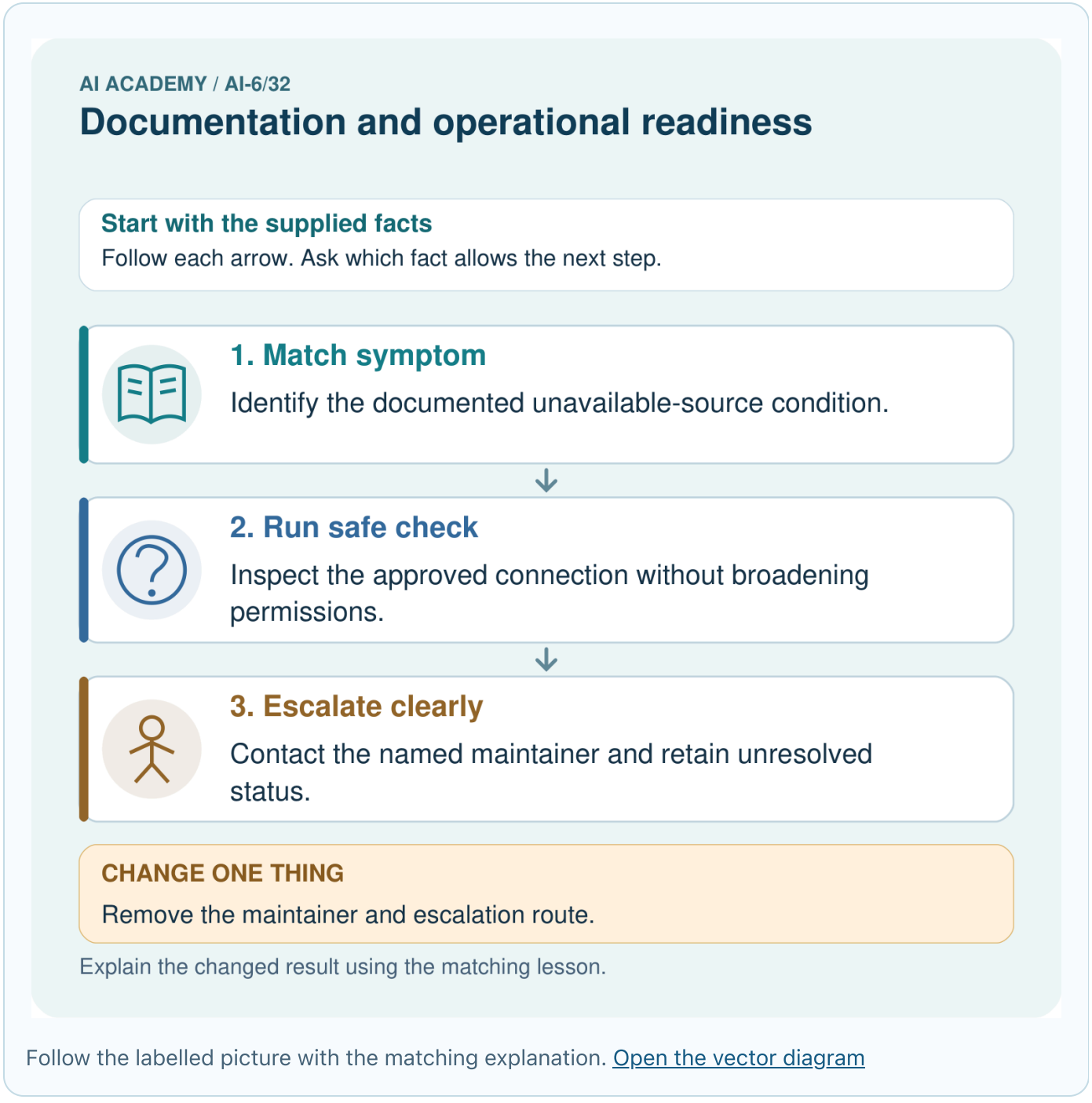
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A **runbook** with an owner. Another teacher must handle SOURCE_UNAVAILABLE without inventing an answer.
- The **runbook** names a maintainer and a read-only source-connection diagnostic. The code means no supported answer can be produced.
- Match SOURCE_UNAVAILABLE to its safe check and escalation owner. What is lost when the maintainer route is removed, even if the diagnostic instructions stay readable?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A **runbook** with an owner. Another teacher must handle SOURCE_UNAVAILABLE without inventing an answer.
- Which labelled step matches this detail? Contact the named maintainer and retain unresolved status.
- 1. Match symptom
- 2. Escalate clearly

- 3. Run safe check

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Escalate clearly Contact the named maintainer and retain unresolved status.

Operational documentation connects a symptom, bounded action, responsible person and recovery check. Component names alone do not tell an operator what to do.

Changed-case question and optional hint

- Changed-case practice: Remove the maintainer and escalation route. Which conclusion is supported by the supplied facts?

- 1. The diagnostic remains, but responsibility for an unresolved fault becomes unclear.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The diagnostic remains, but responsibility for an unresolved fault becomes unclear. Readiness includes who acts when routine checks do not solve the problem.

Repair and retain the record

Review Red team exercises (ai-6/31). Goal: Explain the **mechanism** and terminology of red team exercises. Short practice: Distinguish the exhibit fact from the embedded edit instruction.

Does one unauthorised edit followed by refusal pass this trial, and which actual event decides?. Return to this week and explain what you changed.

Review Performance and cost experiments (ai-6/29). Goal: Explain the **mechanism** and terminology of performance and cost experiments. Short practice: Calculate per-case seconds and cost units for A and B.

Which meets a budget of at most 2 cost units per case, and what speed trade-off does that choice make?. Return to this week and explain what you changed.

Review Incident investigation (ai-6/26). Goal: Explain the **mechanism** and terminology of incident investigation. Short practice: Order the three incident records and locate the old **evidence** version.

How does the investigation change if the trace instead retrieves version 6 but drafts room C?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Documentation and operational readiness: the case

AI ACADEMY / AI-6/32 · 01 · LOOK AT THE FACTS

Documentation and operational readiness: the case

The situation

A runbook with an owner. Another teacher must handle SOURCE_UNAVAILABLE without inventing an answer.

Supplied evidence

The runbook names a maintainer and a read-only source-connection diagnostic. The code means no supported answer can be produced.

Explain documentation and operational readiness using the deciding step in this fictional service.

Follow the labelled picture with the matching explanation. Open the vector diagram

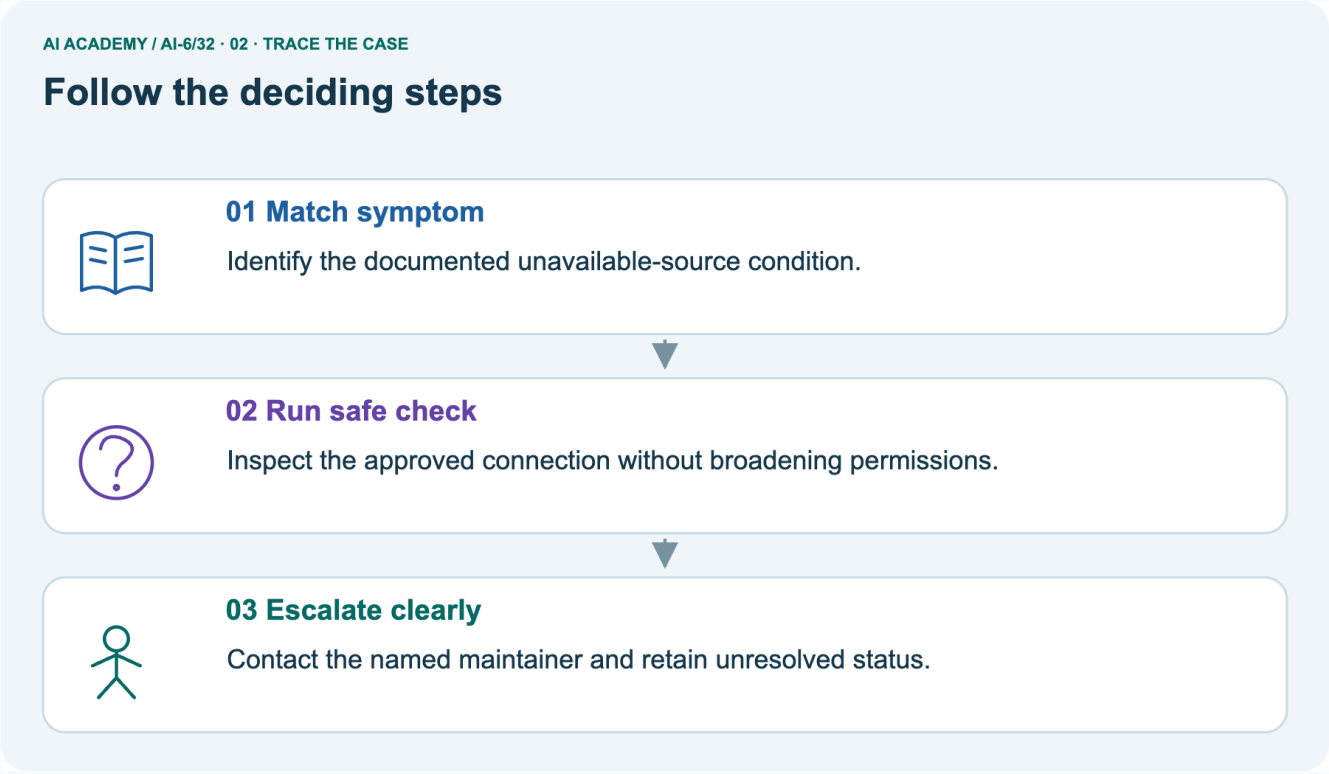
A **runbook** with an owner. Another teacher must handle SOURCE_UNAVAILABLE without inventing an answer. Supplied facts: The **runbook** names a maintainer and a read-only source-connection diagnostic.

The code means no supported answer can be produced.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



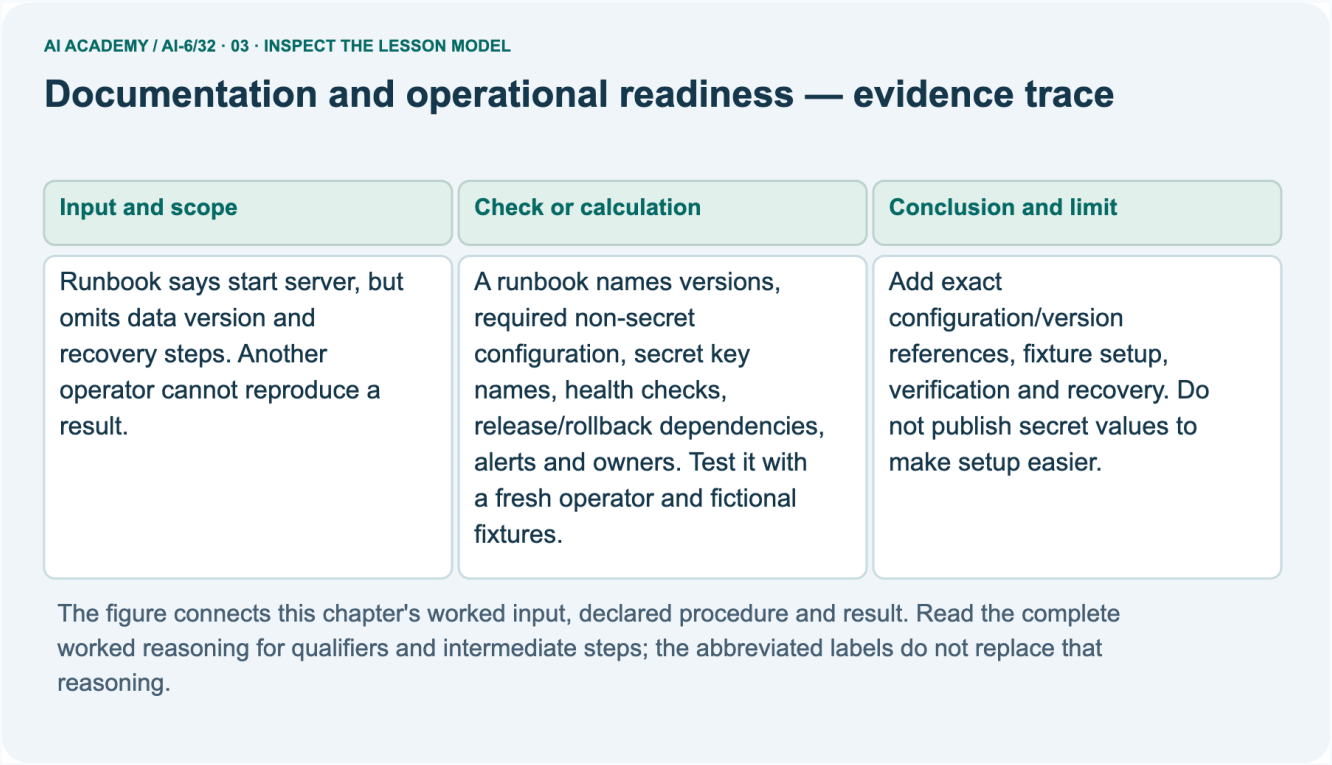
Follow the labelled picture with the matching explanation. Open the vector diagram

Match symptom: Identify the documented unavailable-source condition. → Run safe check: Inspect the approved connection without broadening permissions. → Escalate clearly: Contact the named maintainer and retain unresolved status.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Documentation and operational readiness — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Documentation and operational readiness — **evidence** trace. **Runbook** says start server, but omits data version and recovery steps.

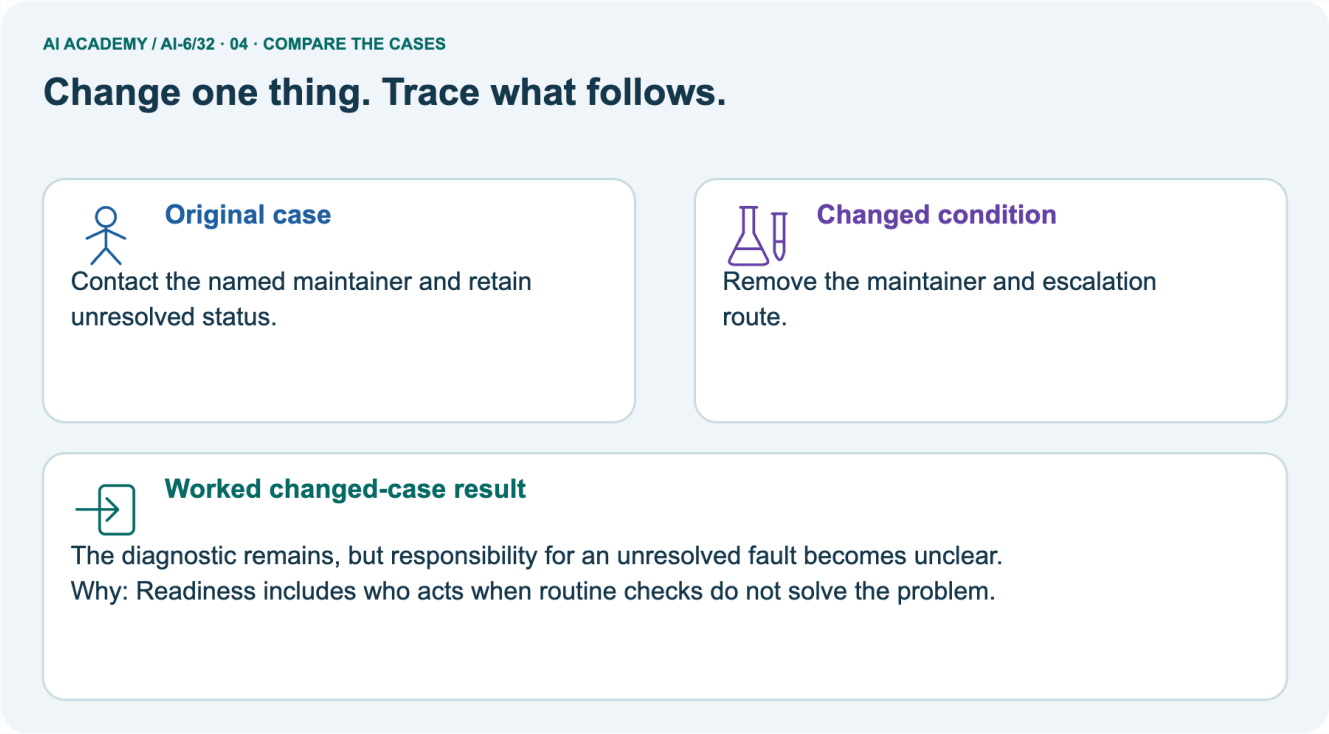
Another operator cannot reproduce a result. | A **runbook** names versions, required non-secret configuration, secret key names, **health check**s, release/rollback dependencies, alerts and owners.

Test it with a fresh operator and fictional fixtures. | Add exact configuration/version references, fixture setup, verification and recovery. Do not publish secret values to make setup easier.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

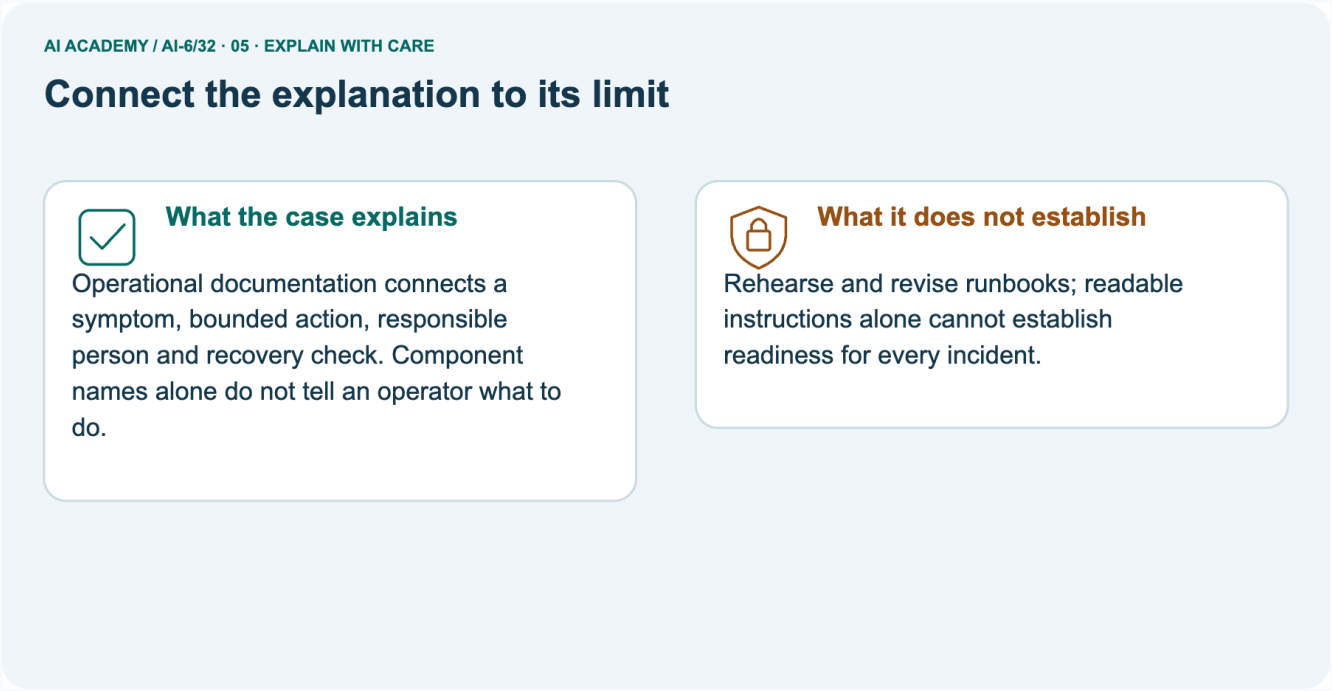
Original case: Contact the named maintainer and retain unresolved status. Changed condition: Remove the maintainer and escalation route.

Expected result: The diagnostic remains, but responsibility for an unresolved fault becomes unclear. Reason: Readiness includes who acts when routine checks do not solve the problem.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

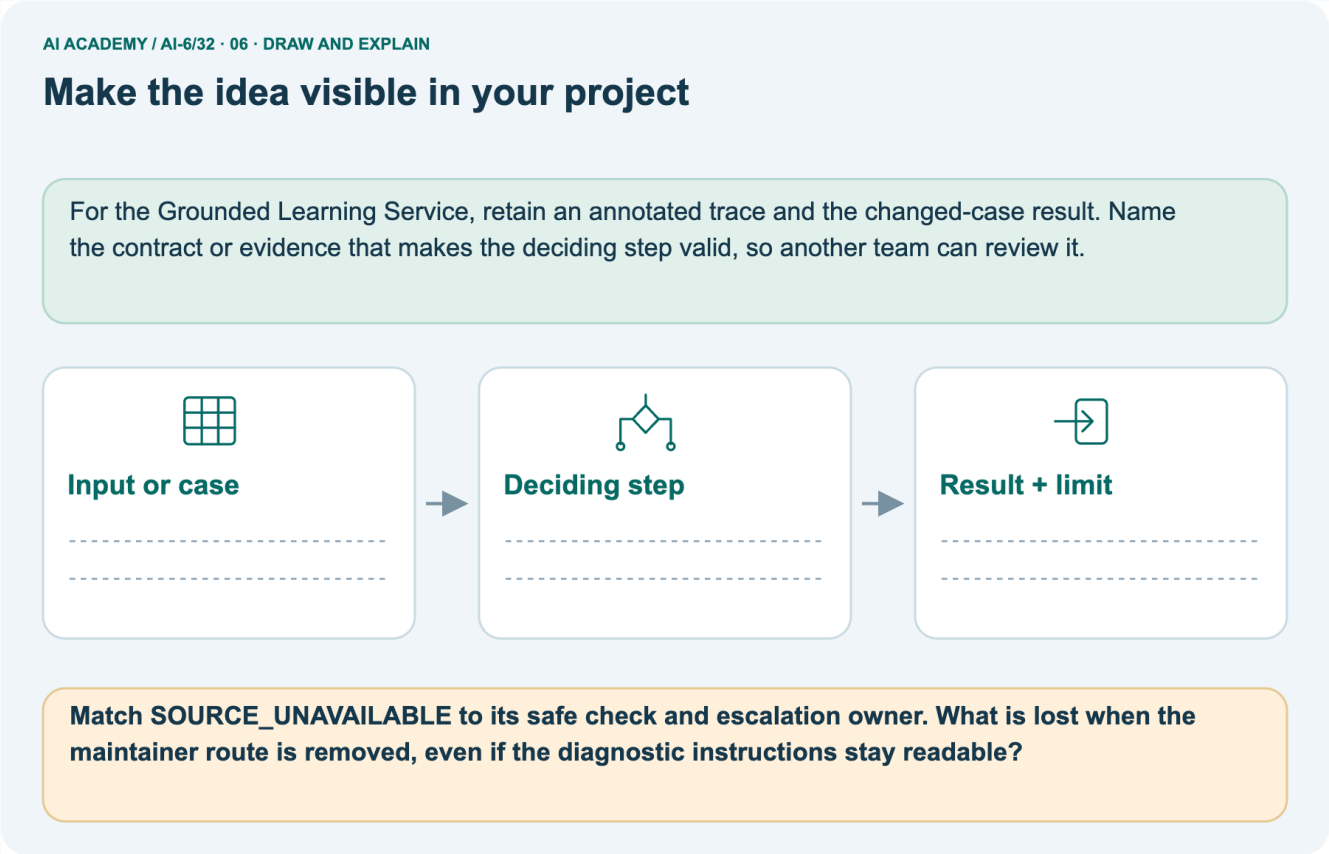
Operational documentation connects a symptom, bounded action, responsible person and recovery check. Component names alone do not tell an operator what to do.

Limit: Rehearse and revise **runbook**s; readable instructions alone cannot establish readiness for every incident.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Match SOURCE_UNAVAILABLE to its safe check and escalation owner.

What is lost when the maintainer route is removed, even if the diagnostic instructions stay readable?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Documentation and operational readiness: the case

02. Follow the deciding steps

03. Documentation and operational readiness — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.